

Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "MacGillivray's Warbler"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

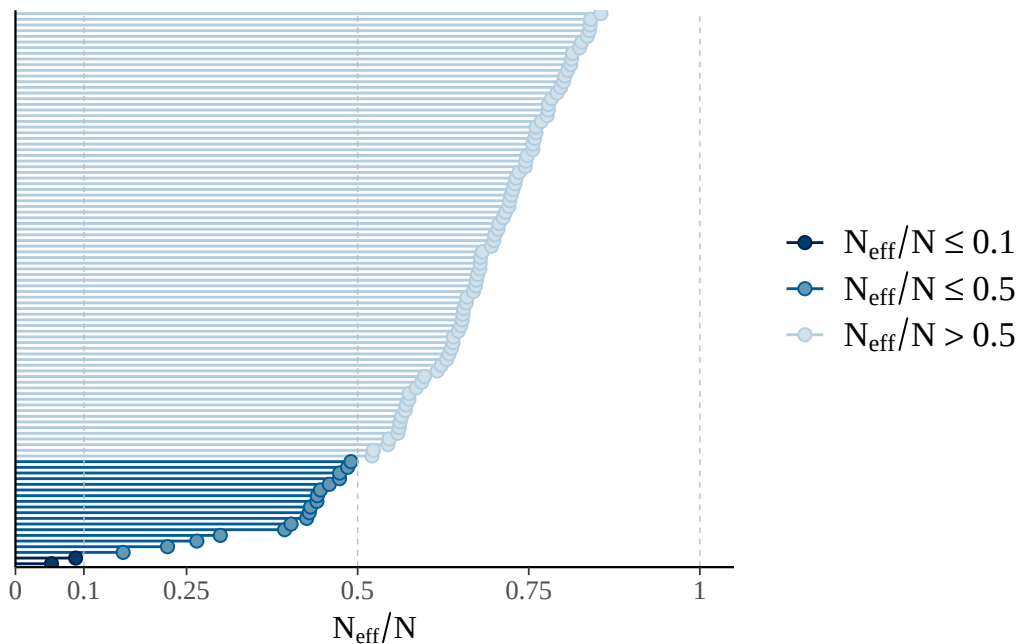
```
[1] "divergent transitions: 0"
```

Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of n_{eff} to N (number of samples), the better.



```
[1] "effective sample size too small"
```

```
[1] "n_eff ratio bulk ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1192	0.6328	0.7876	0.8313	0.9386	1.8863

```
[1] "n_eff ratio tail ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.05272	0.60446	0.69410	0.66748	0.75658	0.87377

```
[1] "effective sample size in bulk or tail too small"
```

Do chains mix well:

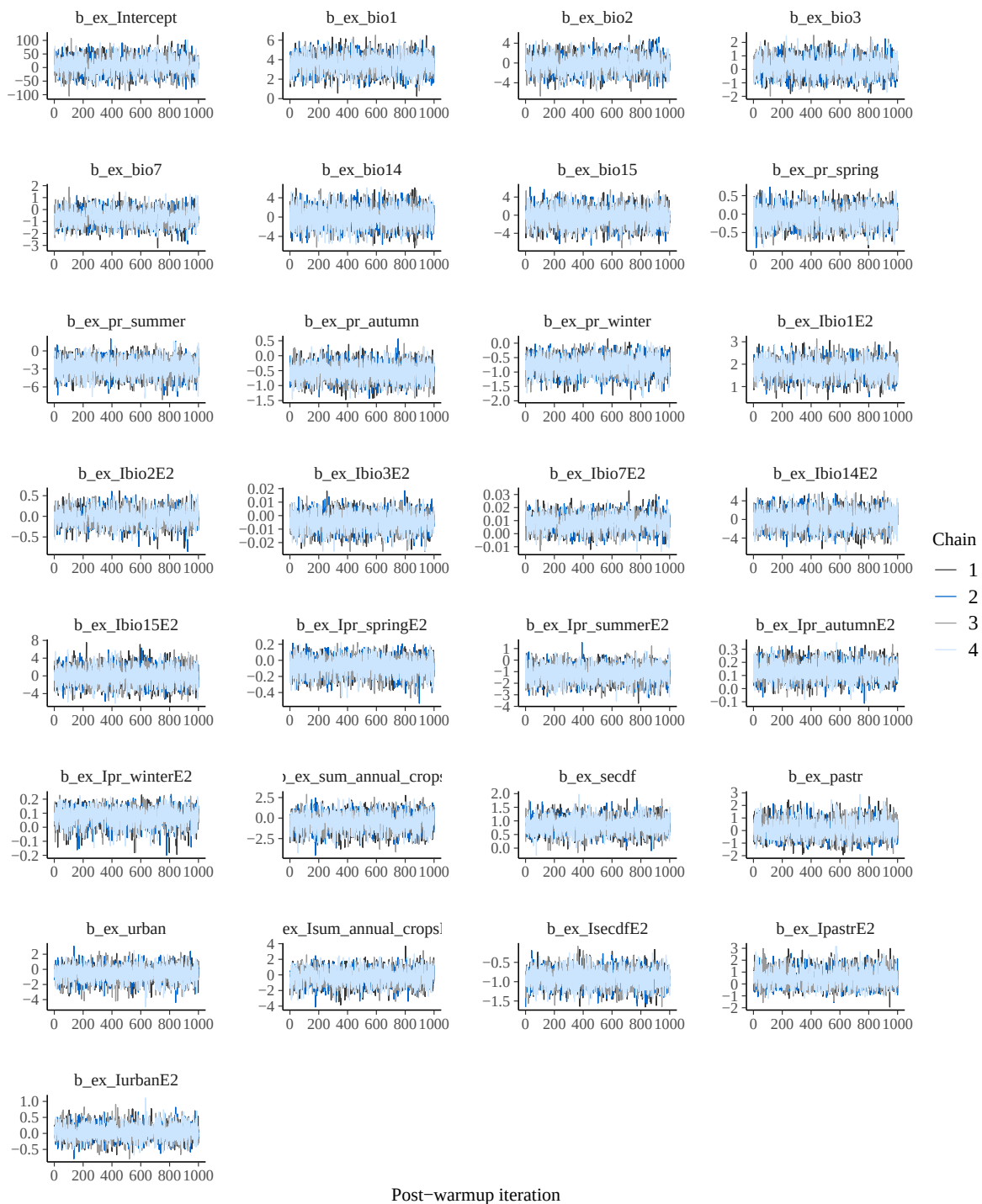
No divergences to plot.

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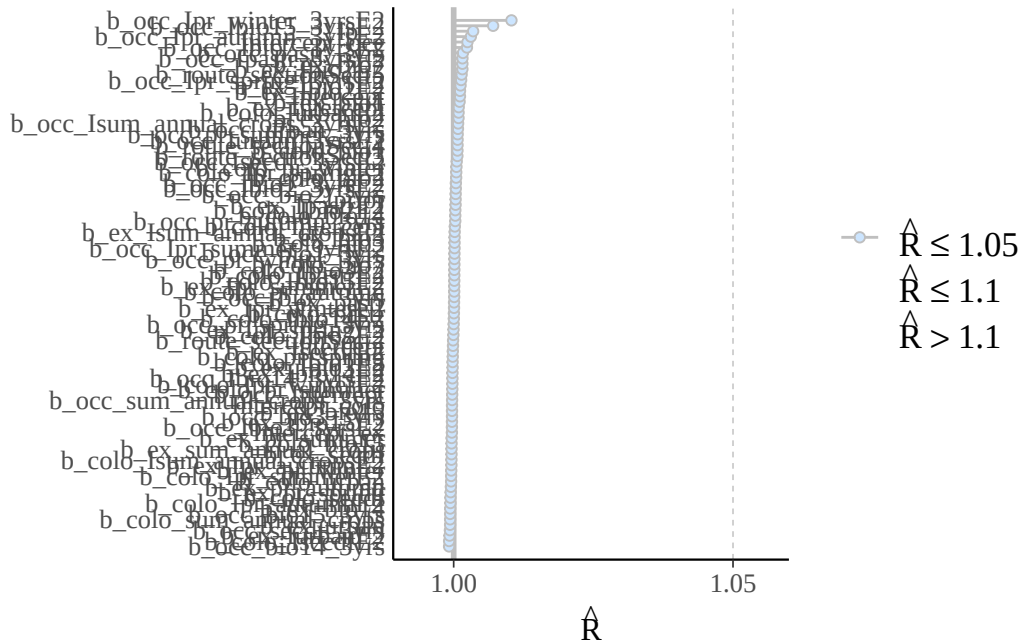
No divergences to plot.





Are R-hat values fine:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9992	0.9997	1.0001	1.0004	1.0007	1.0103



Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_bio1_3yrs"      "b_occ_lbio7_3yrsE2"
```

```
[1] "b_colo_bio1" "b_colo_Ibio7E2"
```

```
[3] "b_colo_Ipr_springE2"      "b_colo_Ipr_winterE2"
```

```
[5] "b_colo_sum_annual_crops"    "b_colo_secdf"
```

```
[7] "b_colo_Isum_annual_cropsE2" "b_colo_IsecdfE2"
```

```
[1] "b_ex_bio1"      "b_ex_pr_summer" "b_ex_pr_winter"
```

```
[4] "b_ex_Ibio1E2"      "b_ex_Ipr_autumnE2" "b_ex_secdf"
```

```
[7] "b_ex_IsecdfE2"
```

